Chapter 2 exercises

1. Use a full truth table to determine whether $((p \to q) \land (q \to r)) \to (p \to r)$ is a tautology. Explain every critical row.
2. Determine whether $\neg p, p \lor q, q \to r \therefore r$ is valid. Give inference rules if valid, or a counterexample valuation otherwise.
3. Put $\neg\exists x \forall y (P(x, y) \land (x < y))$ into negation form with all quantifiers in front. State its domain precisely in words.
4. Prove that if $3n + 2$ is odd, then n is odd. Name your proof method.
5. Find an integer n refuting "if n^2 has remainder one modulo four, then n is prime." Explain how it meets the definition of a counterexample.
6. Over the real numbers, determine the truth of $\forall x \exists y (x + y = 0)$ and $\exists y \forall x (x + y = 0)$. Explain witnesses and their dependence.
7. Simplify $\neg((p \lor q) \land \neg r) \lor q$ using only logical laws, naming the law at every step.
8. Over a nonempty domain, prove $\forall x (P(x) \to Q(x)), \exists x P(x) \therefore \exists x Q(x)$. Mark universal specification, modus ponens, and existential generalization.
9. Does $\forall x (P(x) \lor Q(x)) \to (\forall x P(x) \lor \forall x Q(x))$ hold over integers? If not, provide predicates in one common counterexample interpretation.
10. Prove that $\sqrt{2}$ is irrational. State the contradiction assumption, reduced-fraction condition, each use of the parity lemma, and the final contradiction.
11. Why cannot $\exists x P(x)$ and $\exists x Q(x)$ share one fresh name c to infer $\exists x (P(x) \land Q(x))$? Give a counterexample on one domain and state the conclusion restriction for existential elimination.
12. Give the full truth table of $\varphi = (p \leftrightarrow q) \lor r$, then its canonical DNF and CNF. Express $p \lor q$ with NAND alone, and explain how a general DNF yields a NAND formula for any Boolean function of $n \geq 1$ variables.

References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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https://github.com/xinhi6716/computer-science-self-study